Verify Important Outputs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Verify Important Outputs before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Prompts Are Design Instructions

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Privacy by Choosing Less

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Good Data, Bad Data

Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Verify Important Outputs trace and record the first step you can explain without help.

Use W2: Repair a **unit** error for Verifying **claim**s and quantities. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Verify Important Outputs explanation against the supplied facts before discussing it.

Use the worked case for Verifying **claim**s and quantities. Proposed explanation: “A **citation** proves every nearby **claim**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Audit a new model with mass units

Fictional craft card M states five sealed bags of modelling material at exactly 250 g per bag. Three models each use 300 g, with no other loss.

A draft says “M supplies 1250 g; the models use 900 g; 350 kg remains; every model will win a prize.” Check all four **claim**s, show the remaining mass in grams, and explain the status of the prize **claim**. Use 1 kg = 1000 g.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Verify Important Outputs target to a trace and a justified conclusion.

A generated report says 18 of 24 tests passed, so success was **80%**. Verify the **claim** in two ways and write a corrected sentence.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Verify Important Outputs case and show the deciding evidence.

An answer says six groups of four make 28.

Verify **independent**ly.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

An assistant says the school library opens at 8, but today's official notice says 9. Which **source** should guide today's visit? Explain the check.

My **independent** answer and reason

> **Write or draw your reasoning and evidence.**

Week 27 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Verify Important Outputs adds to Helpful Classroom Sorter and keep the evidence.

Require human confirmation for important sorter **output**s and attach the **evidence** record used to approve each rule or label.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A generated report says 18 of 24 tests passed, so success was **80%**. Verify the **claim** in two ways and write a corrected sentence.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Separate source facts from derived quantities

> Separate source facts from derived quantities. Apply the supplied Verify Important Outputs facts and keep your reasoning.

Using R, identify a directly stated fact and a quantity that must be calculated. Give the **source** or expression for each. Explain why a **citation** to R does not verify the **claim** that every participant will enjoy the activity.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Repair a unit error

> Repair a unit error. Apply the supplied Verify Important Outputs facts and keep your reasoning.

A report calculates 12 × 50 = 600 and labels the result metres. Use R to correct the **unit**, convert the result to metres and calculate the remaining length.

Give a second arithmetic route or a check using starting, used and remaining lengths.

> **My reasoning and evidence**

W3 Do not count copied claims as separate evidence

> Do not count copied claims as separate evidence. Apply the supplied Verify Important Outputs facts and keep your reasoning.

Three posters say 600 metres were used, but all copied one draft. A fourth poster cites card Z, which is unavailable. Explain why these posters do not settle the quantity.

Name the available **evidence** and check you would use instead.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Audit a new model with mass units

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Audit a new model with mass units. Apply the supplied Verify Important Outputs facts and keep your reasoning.

Fictional craft card M states five sealed bags of modelling material at exactly 250 g per bag. Three models each use 300 g, with no other loss.

A draft says “M supplies 1250 g; the models use 900 g; 350 kg remains; every model will win a prize.” Check all four **claim**s, show the remaining mass in grams, and explain the status of the prize **claim**. Use 1 kg = 1000 g.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 State what a missing source prevents you from knowing

> State what a missing source prevents you from knowing. Apply the supplied Verify Important Outputs facts and keep your reasoning.

Another draft **claim**s that the amount left in W4 accounts for all real-world waste, citing a missing instruction card. Explain which result can still be checked from M and which **claim** remains unresolved.

What information would be needed before revising the no-loss model?

> **My reasoning and evidence**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

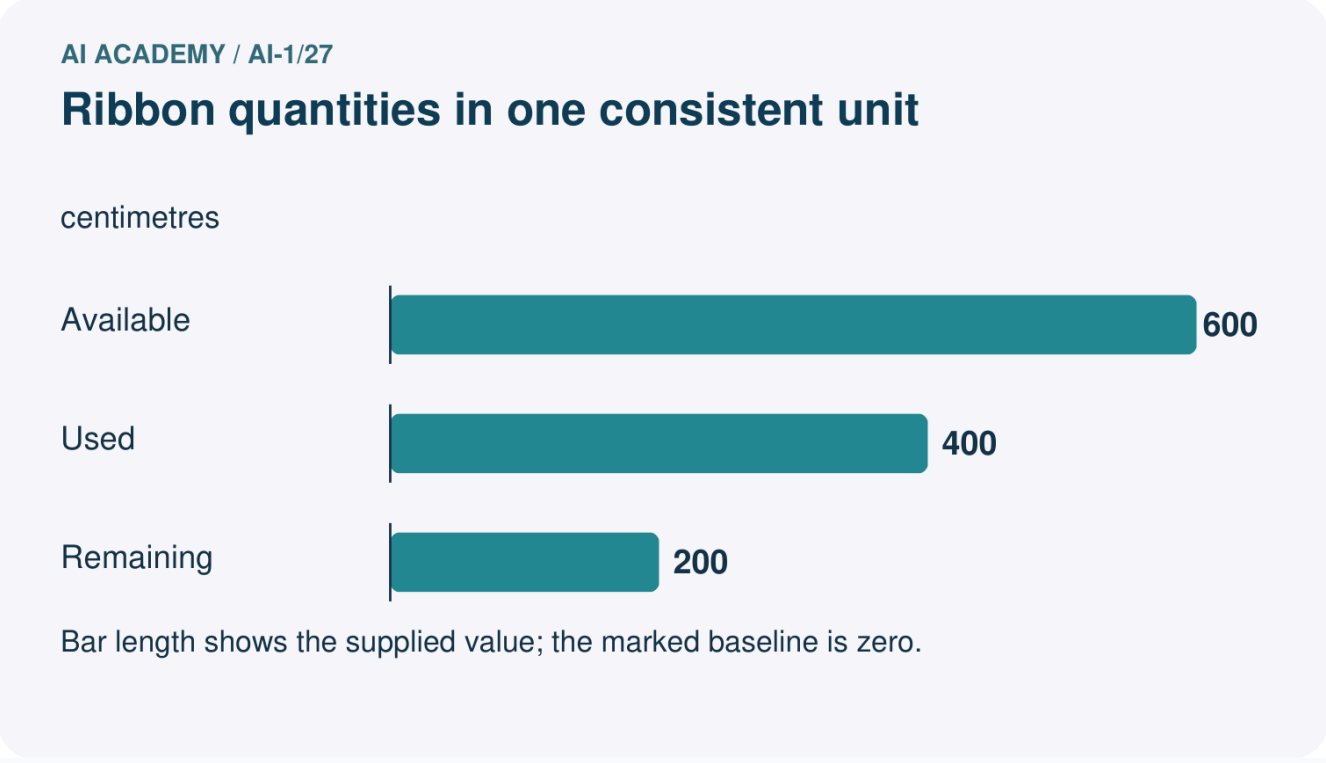
Pictorial case practice: Verify Important Outputs

> A fictional craft sheet calculates ribbon left after making bookmarks. Explain the labelled step before checking the changed condition.

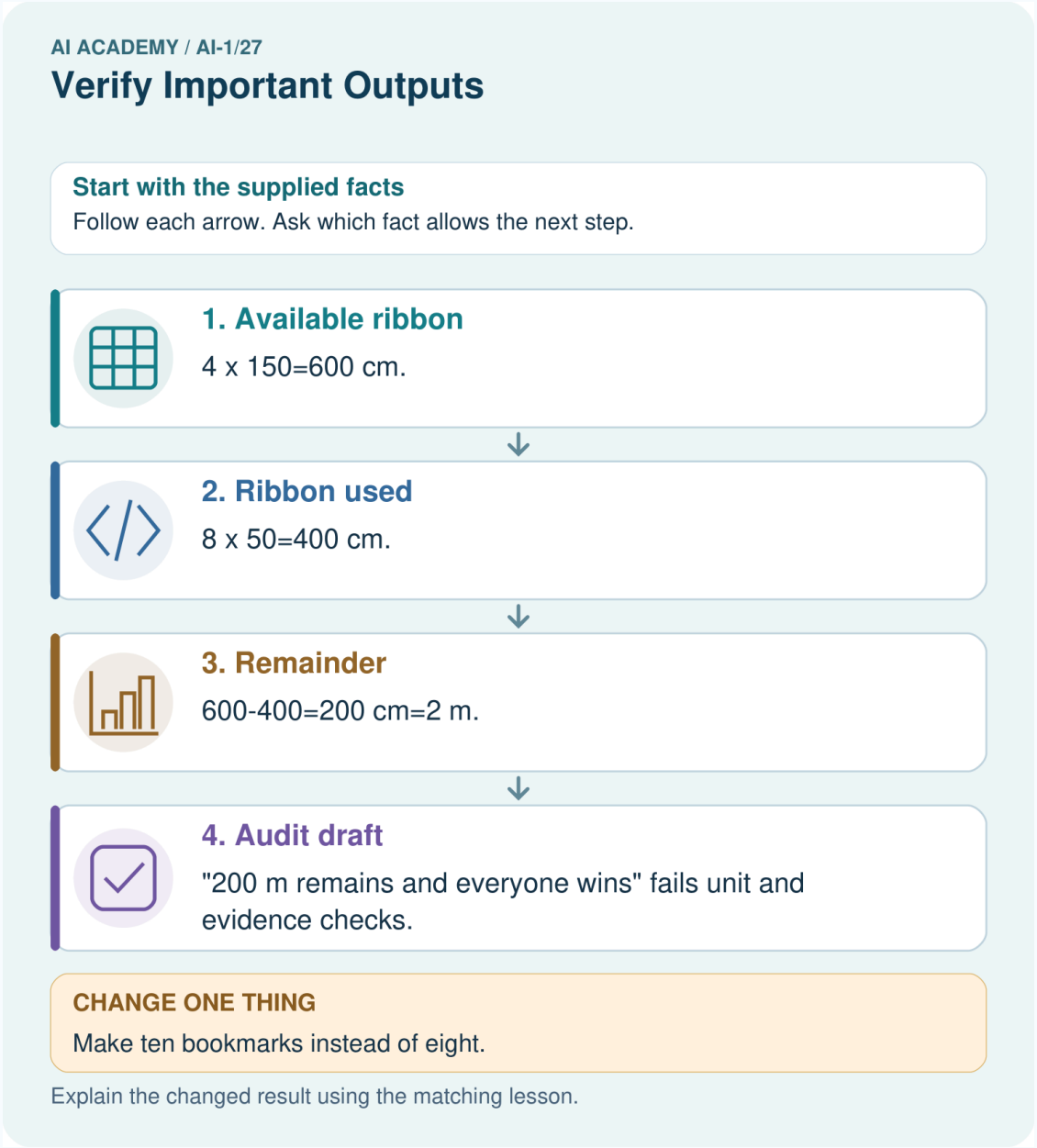
A fictional craft sheet calculates ribbon left after making bookmarks.

Four rolls each contain exactly 150 cm. Eight bookmarks use exactly 50 cm each.

No other ribbon loss is assumed; no competition result is recorded.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Correct the draft and identify which part needs another **source** rather than a calculation.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Check sorter calculations with explicit **unit**s and verify explanation **claim**s against their actual catalogue **evidence**.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Check sorter calculations with explicit **unit**s and verify explanation **claim**s against their actual catalogue **evidence**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved **independent** assessment for a separate first attempt before teacher feedback.

A fictional craft sheet calculates ribbon left after making bookmarks.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? "200 m remains and everyone wins" fails **unit** and **evidence** checks.

1. Audit draft

2. Remainder

3. Ribbon used

4. Available ribbon

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Audit draft “200 m remains and everyone wins” fails **unit** and **evidence** checks. All arithmetic first uses centimetres.

Converting only after subtraction avoids mixing **unit**s. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning.

A numerically correct total elsewhere in a draft cannot rescue a wrong **unit** or an invented **claim**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Make ten bookmarks instead of eight. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. 100 cm, or 1 m, remains.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: 100 cm, or 1 m, remains.

600−10×50=100 cm.

Repair a missing building block

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements.

Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields.

Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Review Good Data, Bad Data (ai-1/21). Goal: Check records against a stated format, **unit** and meaning before using them. Short practice: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

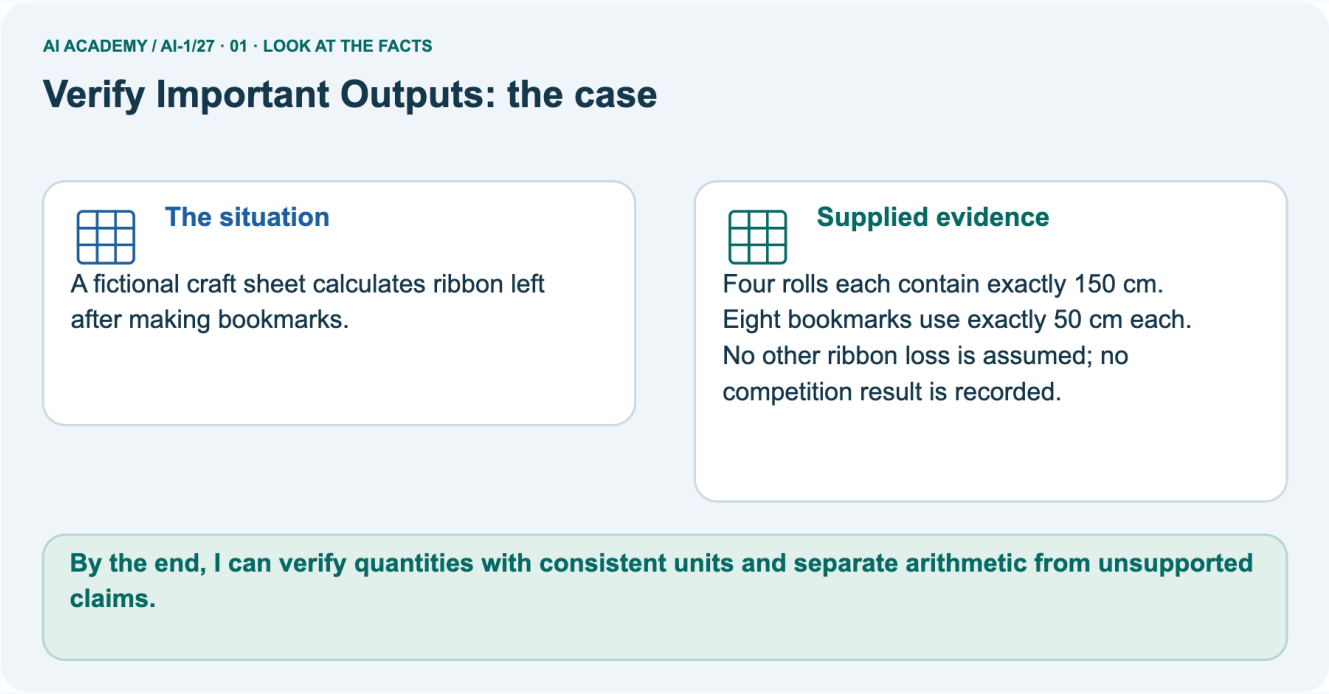
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Verify Important Outputs picture from its labels, then record a relevant limit.

01. Verify Important Outputs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

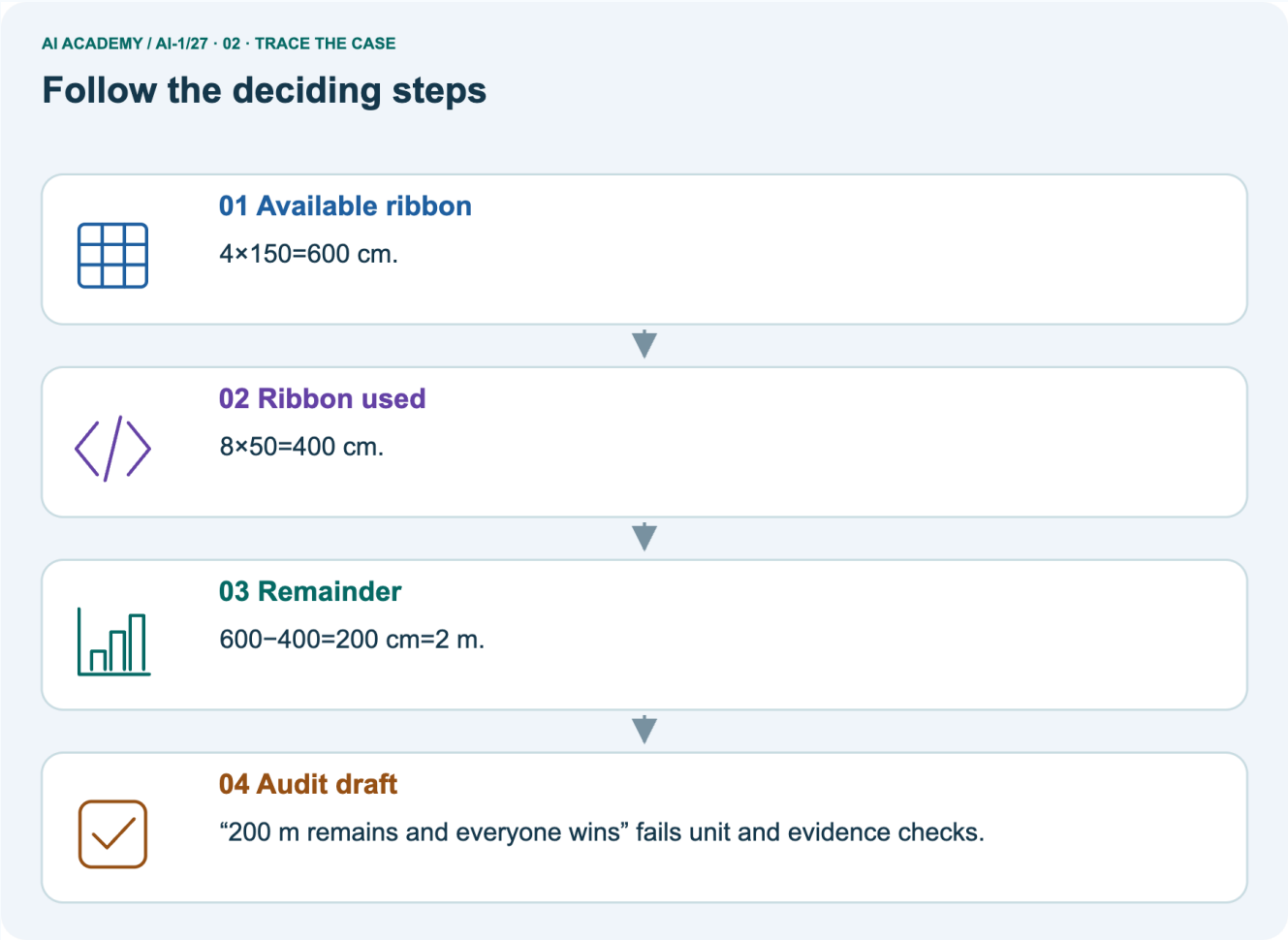
A fictional craft sheet calculates ribbon left after making bookmarks. Supplied facts: Four rolls each contain exactly 150 cm. Eight bookmarks use exactly 50 cm each.

No other ribbon loss is assumed; no competition result is recorded.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Available ribbon: 4×150=600 cm. → Ribbon used: 8×50=400 cm. → Remainder: 600−400=200 cm=2 m. → Audit draft: “200 m remains and everyone wins” fails **unit** and **evidence** checks.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Ribbon quantities in one consistent unit



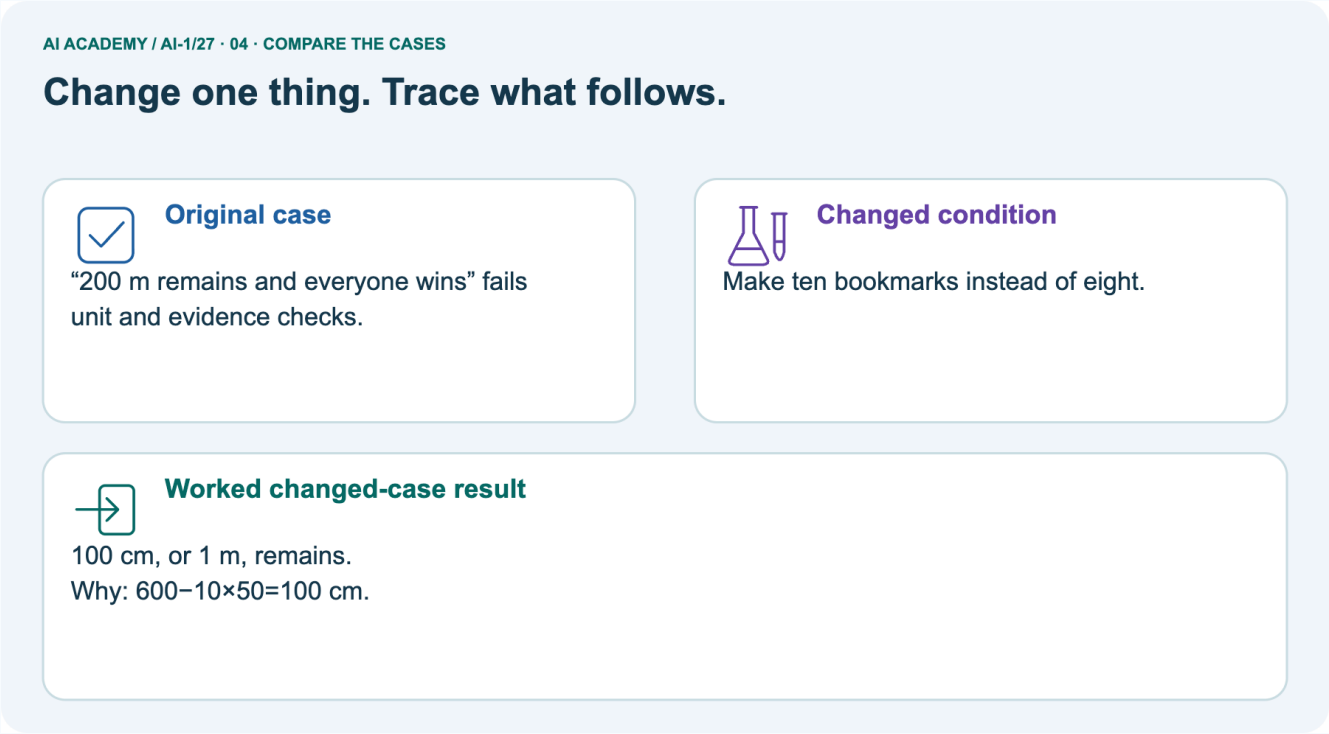
Follow the labelled picture with the matching explanation. Open the vector diagram

Ribbon quantities in one consistent **unit**. **Unit**s: centimetres. Available: 600. Used: 400. Remaining: 200.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



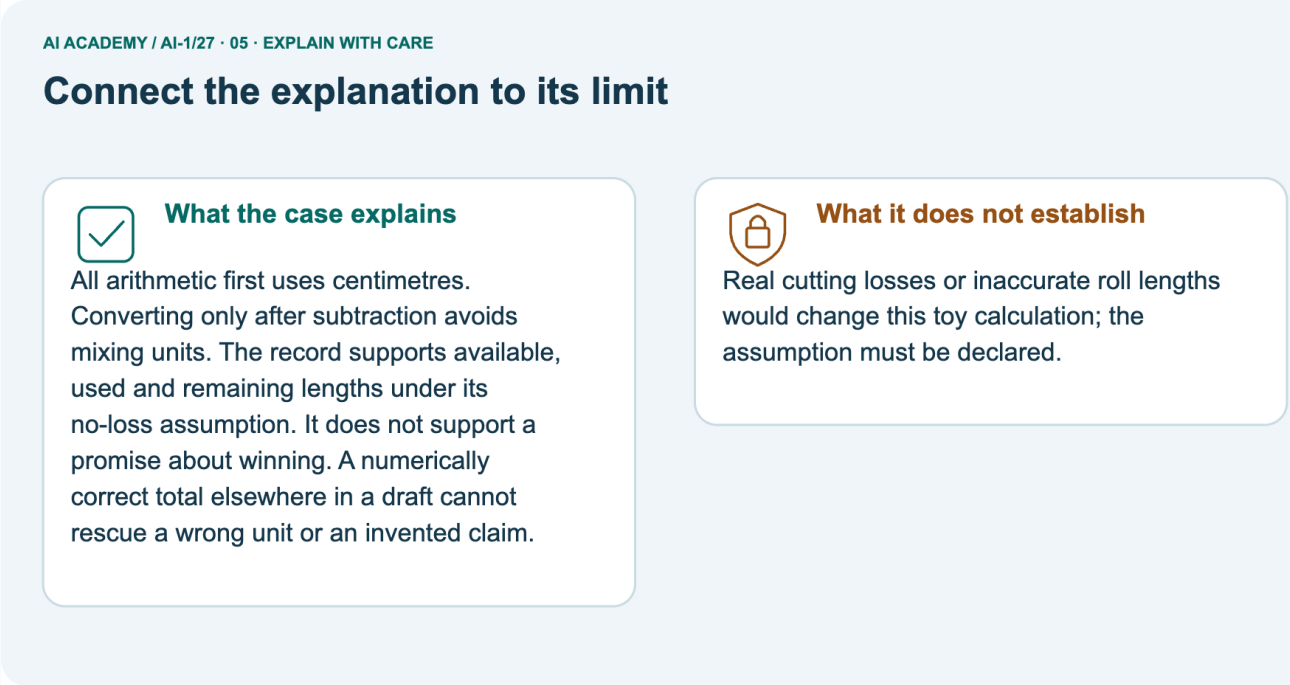
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: “200 m remains and everyone wins” fails **unit** and **evidence** checks. Changed condition: Make ten bookmarks instead of eight. Expected result: 100 cm, or 1 m, remains. Reason: 600−10×50=100 cm.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

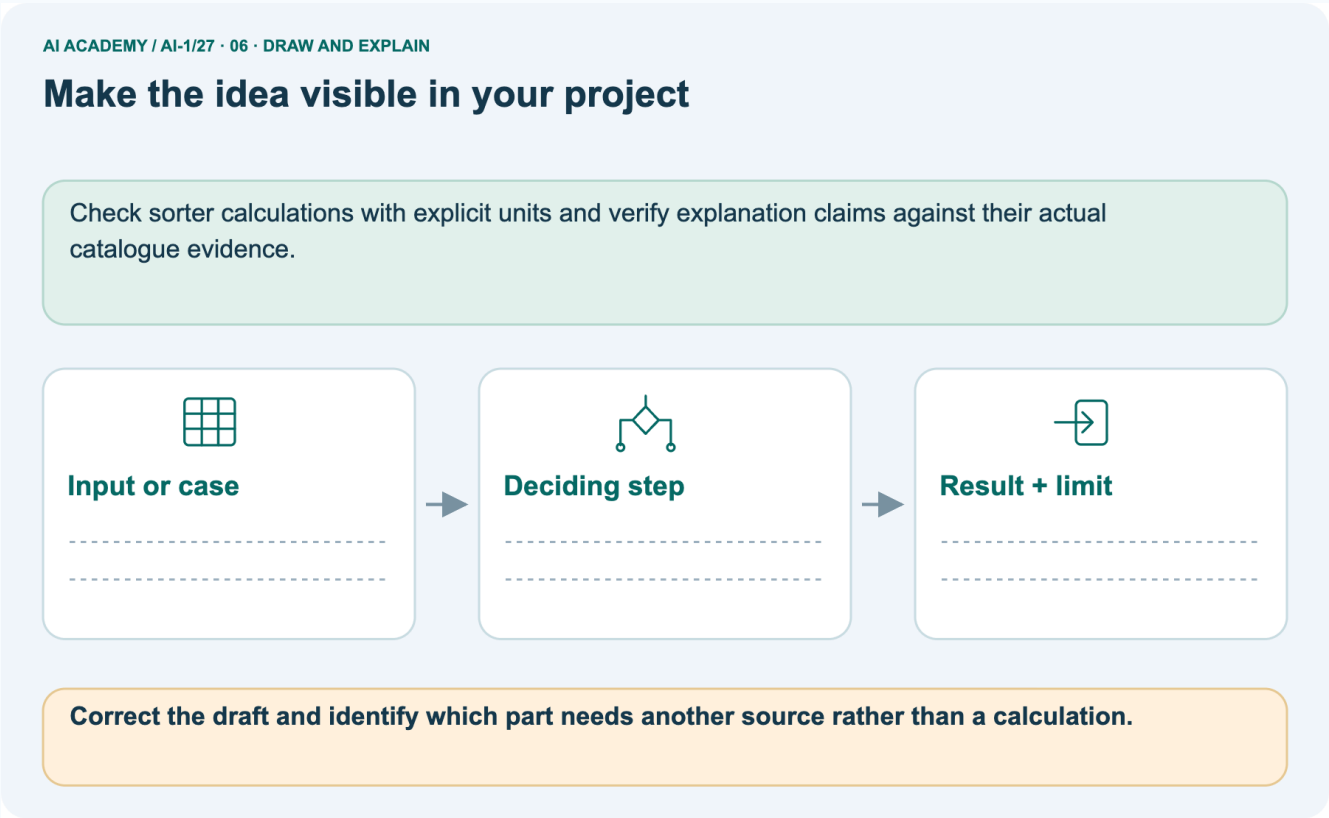
All arithmetic first uses centimetres. Converting only after subtraction avoids mixing **unit**s. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning.

A numerically correct total elsewhere in a draft cannot rescue a wrong **unit** or an invented **claim**. Limit: Real cutting losses or inaccurate roll lengths would change this toy calculation; the assumption must be declared.

Explain the case, then point to a limit on the **claim**.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Check sorter calculations with explicit **unit**s and verify explanation **claim**s against their actual catalogue **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Correct the draft and identify which part needs another **source** rather than a calculation.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.